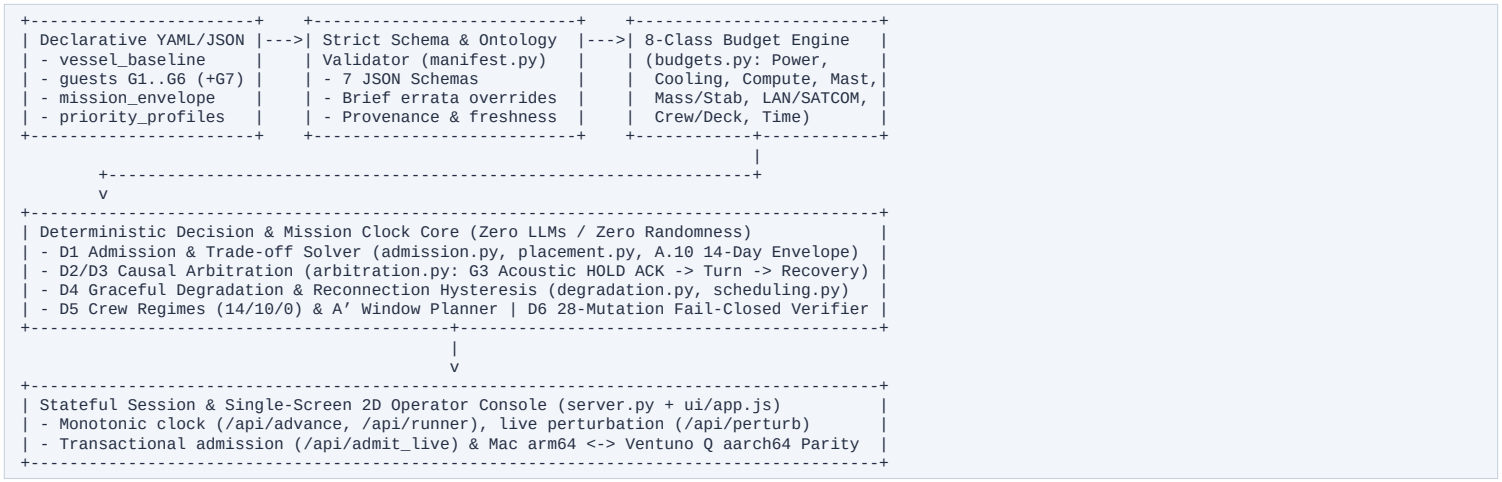
SDV Eight-Class Resource Governor (sdv-resource-governor)

Deterministic, ontology-driven Admission (D1), Reconfiguration (D2), Live Arbitration (D3), Graceful Degradation (D4), Multi-Crew Scheduling (D5), and Fail-Closed Evidence Governance (D6) for a 72 m Offshore Software-Defined Vessel (SDV).

1. Architecture Diagram & Data Flow



2. Deliverables, Objectives & Scope (D1-D6: Mandatory vs. Desirable)

ID	Deliverable / Capability	Brief Tier	Status & Implementation Scope
D1	Guest Manifest Ontology & 7 Formal JSON Schemas	Mandatory	Implemented: schemas/*.schema.json, G1-G6 YAML cards + dynamic G7 card, bidirectional requires_guests / excludes_guests / conditional_excludes.
D2	Port Admission, Placement & Trade-off Solver (Scenario A)	Mandatory	Implemented: Exact 8-class + 14-day Mission Envelope A.10 solver (YES / YES_IF / NOT_NOW / NO), ranked displacements, atomic apply_yes_if commit gate.
D3	Live Conflict Arbitration & Causal Recovery (Scenario B)	Mandatory	Implemented: Causal state machine (HOLD_CMD -> HOLD_ACK -> TURN_INTO_WIND -> RECOVERY_APPROACH -> TOUCHDOWN <= 1200 s) under survey_primary & rescue_primary.
D4	Graceful Degradation, Thermal Inertia & Hysteresis	Mandatory	Implemented: 100 kW generator loss & 30 deg C / -40% cooling failure mitigation, 60 s thermal inertia (1.8 MJ), 60 s cooldown + 15 kW reconnection hysteresis.
D5	Single-Screen 2D Operator Console & Live What-If	Mandatory	Implemented: Zero-build 2D SVG/HTML/JS console (http://127.0.0.1:8765) bound to stateful GovernorSession (/api/advance, /api/perturb, /api/admit_live).
D6	Scenario C (A', Multi-Crew 14/10/0, Lost Feeds) & Audit	Desirable	Implemented (Real Scope): Quantitative 14/10/0 crew table, A' earliest window (T+1260s), +20% overconsumption & stale-feed fallback, 28-mutation verifier, and Mac arm64 <-> Ventuno Q aarch64 parity.

3. Clean / Offline Installation & Quickstart (<= 5 minutes)

```
# 1. Create clean virtual environment (offline wheelhouse or system site-packages supported)
python3 -m venv .venv --system-site-packages
.venv/bin/pip install -r requirements-macos-arm64.lock
.venv/bin/pip install --no-build-isolation --no-deps -e .

# 2. Run the full automated test suite (Mac arm64 or Linux aarch64)
.venv/bin/pytest -v

# 3. Generate deterministic campaign bundle & run fail-closed verifier + 28 adversarial mutations
.venv/bin/sdv-governor campaign --run-id demo_run --junit-xml artifacts/final_run/junit.xml
.venv/bin/sdv-governor verify-evidence artifacts/demo_run --run-mutations

# 4. Launch the interactive single-screen 2D Operator Console (Ctrl+C to stop)
.venv/bin/sdv-governor serve --host 127.0.0.1 --port 8765
```

4. Eight-Class Resource Model (PHY-01..PHY-10)

Class	Physical / Operational Constraint	Source Module
1. Power	200 kW reserve (LINE_A_FWD <= 110 kW, LINE_B_AFT <= 130 kW), peak vs. sustained, firm reservations	budgets.py
2. Cooling	Chilled water curve 160 kW @ <=12 deg C -> 120 kW @ 30 deg C (-40% loop fault), 1:1 heat, 30 kW x 60 s inertia	budgets.py
3. Compute	Rack node_1 (16c/64G/1 GPU), node_2 (16c/64G/1 GPU), node_3 (8c/32G/0 GPU)	placement.py
4. Mast	mast_3 single-slot exclusivity, <= 60 kg, <= 0.6 m^2, <= 2,000 W mast air cooling	budgets.py
5. Mass & Stability	Global <= 10,000 kg and <= 12,000 kg-eq (mass_kg x height_factor); bay1 <= 8t, bay2 <= 4t	budgets.py
6. Data Links	10 Gbps LAN (<= 1 Gbps/port), SATCOM 8 Mbps port / 6 Mbps offshore / 1.5 Mbps degraded, EMCON	budgets.py
7. Crew & Deck	Regimes 14 (9 avail), 10 (5 avail), 0 (USV, 0 avail); flight_deck vs stern_ramp exclusivity	scheduling.py
8. Time	1 Hz tick budget (<= 1,000 ms), Scenario A admission (<= 30 s), local urgent command (<= 200 ms)	replay.py

5. Adding a New Guest (G7) Without Touching Code (EXT-01)

Any new guest payload is admitted purely from a declarative dictionary or YAML file:

```
import yaml
from sdv_governor.manifest import parse_guest_manifest

raw = yaml.safe_load("""
schema_version: "sdv-guest-manifest/1.0"
guest_id: "G7"
name: "G7 - Compact Expendable Bathythermograph (XBT) Launcher"
category: "AUXILIARY_PAYLOAD"
safety_critical: false
default_mode: "STANDBY"
```

```
can_buffer_offline: true
subcomponents:
  - {subcomponent_id: "G7_DECK_LAUNCHER", part_of: "G7", name: "Stern XBT Pod", mass_kg: 180,
     footprint_m2: 1.2, wind_area_m2: 0.8, preferred_positions: ["bay1_b", "bay2"],
     allowed_positions: ["bay1_b", "bay2", "bay1_a"], cooling_route: "CHILLED_WATER_LOOP",
     requires_water_manifold: false, value_origin: "ASSUMED"}
modes:
  STANDBY:
    {description: "Launcher indexed and heater idle", min_power_w: 200, nominal_power_w: 300,
     sustained_power_w: 300, sustained_duration_s: 3600, peak_power_w: 500, peak_duration_s: 10,
     cooling_water_w: 300, cooling_air_mast_w: 0, compute_cores: 1, compute_ram_mib: 1024,
     compute_accelerators: 0, shared_storage_gb: 20, backbone_lan_bps: 1000000, satcom_bps: 0,
     safety_critical: false, uninterruptible: false, min_done_duration_s: 0, min_cooldown_s: 0,
     forbids_emcon_silence: false, requires_deck_clear: false, requires_quiet_acoustic_state: false,
     capability_retention_ratio: 0.25, value_origin: "ASSUMED"}
installation: {port_only: true, min_crew_required: 2, install_time_min: 30, remove_time_min: 20,
               sea_install_max_wind_kn: 0, sea_install_min_speed_kn: 0.0, sea_install_max_speed_kn: 0.0}
vehicle_replacement: {supported_at_sea: false, min_crew_required: 0, swap_duration_min: 0,
                      min_speed_kn: 0.0, max_speed_kn: 0.0, max_wind_kn: 0, max_sea_state: 0}
ship_state_constraints: {min_speed_kn_for_operation: 0.0, max_speed_kn_for_operation: 20.0,
  max_wind_kn_for_operation: 35, max_sea_state_for_operation: 6,
  requires_heading_into_wind_for_launch_recovery: false, forbids_deck_machinery_while_active: false,
  acoustic_hold_handshake_min_s: 0, acoustic_hold_handshake_max_s: 0, acoustic_hold_default_s: 0,
  low_battery_threshold_pct: null, emergency_landing_deadline_s: null, max_flight_endurance_min: null}
dependencies: {requires_guests: [], excludes_guests: [], conditional_excludes: []}
mission_capabilities: {hydro_sampling_value: 20.0}
""")
manifest = parse_guest_manifest(raw)
```

6. Explicit Limitations & Honest Operational Boundaries

* 1. Software Dispatch vs. Physical Ship Actuation (< 1 ms vs. Physical Plant): The < 1 ms (~0.015 ms on Mac arm64, ~0.032 ms on Ventuno Q aarch64) latency measured for URGENT_STOP / HOLD / SECURE reflects local Python governor state transition and command emission only (<= 200 ms software budget). Real vessel breaker opening, hydraulic winch braking, acoustic modem propagation (10 s nominal delay modeled in D3), and vessel heading stabilization (90 s modeled in D3) are physical plant processes governed by external PLC/PMS hardware.

* 2. Discrete 20 s Simulation Resolution & Linearized Plant Curves: Scenario B executes at 20 s discrete ticks (90 steps over 1800 s) with piecewise-linear seawater cooling capacity (160 kW -> 120 kW) and a single-node lumped thermal inertia reservoir (1.8 MJ, 600 s cooldown) rather than a continuous CFD/pipe-network solver.

* 3. Combinatorial Displacement Horizon: Port trade-off search bounds simultaneous displacements to k <= 3 installed guests (O(N^3)), which is exact for naval modular bays (N <= 10) without requiring an external MILP solver.

* 4. Hardware Peripheral Scope: STM32H5 MCU real-time motor control is not required by the naval governor challenge; Ventuno Q (training06, aarch64) executes the full Python governor and verification suite over LAN (ssh ardu).

7. What We Would Do With One More Week

* 1. Closed-LoopHIL Bus Integration (Modbus TCP / NMEA 2000 / CAN): Connect FeedReading telemetry and pending_commands directly to a hardware-in-the-loop PMS/winch simulator over NMEA 2000 and Modbus TCP with hardware watchdog heartbeats on the Ventuno Q STM32H5 co-processor.

* 2. Continuous Multi-Compartment Thermal & Power Transient Model: Replace single-reservoir thermal inertia with per-bay chilled-water flow/temperature differential equations and generator governor droop curves during 35 kW UAV arrestor peaks.

* 3. Rolling-Horizon Weather & Sea-State Forecast Arbitrator: Extend D3/D5 arbitration from a single 30-minute conflict window to a 72-hour rolling metocean forecast that pre-positions UAV recovery and battery charging before sea-state degradation.

* 4. Cryptographic Signed Audit Ledger: Sign every evidence_manifest.json and live operator override (URGENT_STOP, apply_yes_if) with hardware-backed Ed25519 keys for maritime regulatory black-box compliance.

8. Repository Structure

* src/sdv_governor/ -- Deterministic engine (models.py, manifest.py, budgets.py, placement.py, admission.py, arbitration.py, degradation.py, scheduling.py, replay.py, evidence.py, server.py, cli.py, pdf_export.py).

* data/ & schemas/ -- Vessel baseline (vessel_baseline.yaml, mission_envelope_a10.yaml), guest manifests (G1-G6), priority profiles, scenarios, errata, and 7 JSON Schemas (mirrored in src/sdv_governor/_bundled/).

* ui/ -- Single-screen 2D Operator Console (index.html, styles.css, app.js).

* docs/ -- Submission documentation (EXECUTIVE_SUMMARY.md, TECHNICAL_DOSSIER.md, OPERATOR_PLAYBOOK.md, PITCH_SLIDES.md, DEPLOYMENT_AND_BOM.m
KNOWN_LIMITATIONS.md + PDFs).

* tests/ & artifacts/ -- Unit, integration, and adversarial regression suites + verified release bundles (release_s6, release_s6_patch1).